

Project Documentation

Project Title: Diabetic data analysis using Pandas, Numpy Matplotlib, and Seaborn

1. Project Overview

This project delivers an end-to-end data analytics solution focused on clinical data from diabetic patients. The primary objective is to investigate the demographic, clinical, and administrative factors that influence patient health outcomes and hospital readmission rates. By transforming raw, unverified healthcare records into an analytical dataset, this study uncovers critical utilization patterns, flags high-risk patient segments, and provides data-driven insights to help healthcare providers optimize patient care and reduce preventable readmissions.

2. Tools Used

- **Pandas** – Data cleaning and manipulation
- **Matplotlib** – Basic data visualization
- **Seaborn** – Advanced and statistical visualizations

3. Dataset

- **Source:** Clinical healthcare dataset containing diabetic patient records.
- **Description:** The dataset contains structured clinical information with key features such as:
 - Patient ID / Encounter ID
 - Age Group / Age Cohort
 - Gender
 - Time in Hospital (Days)
 - Number of Lab Procedures
 - Number of Medications

- Primary Diagnosis / Comorbidities
- Readmission Status (Yes / No)

4. Steps Followed

1.Imported the Dataset: Loaded the raw diabetic patient dataset into a Pandas DataFrame.

2.Cleaned and Preprocessed the Data:

- Handled missing values using data imputation techniques (such as mean, median, or mode imputation for clinical markers).
- Handled anomalies and verified that patient records were consistent.
- Converted medical and demographic columns into appropriate data types for analysis.

3. Feature Engineering

- Segmented continuous variables into structured bins (e.g., grouping continuous ages into distinct age cohorts).
- Created binary flags to simplify the analysis of target variables like patient readmission.

4.Performed Exploratory Data Analysis (EDA):

- Used Pandas to evaluate statistical distributions, slice data by demographic groups, and calculate clinical averages.

5.Created Visualizations: Utilized Matplotlib and Seaborn to construct:

- **Histograms & Box Plots** to analyze the distribution of hospital stay lengths and identify outliers.

- **Bar Charts & Count Plots** to compare readmission rates across different age cohorts and genders.
- **Correlation Heatmaps** to inspect the relationships between lab procedures, number of medications, and time spent in the hospital.
- **Interpreted Results:** Analyzed the generated plots and statistical summaries to extract meaningful clinical insights.

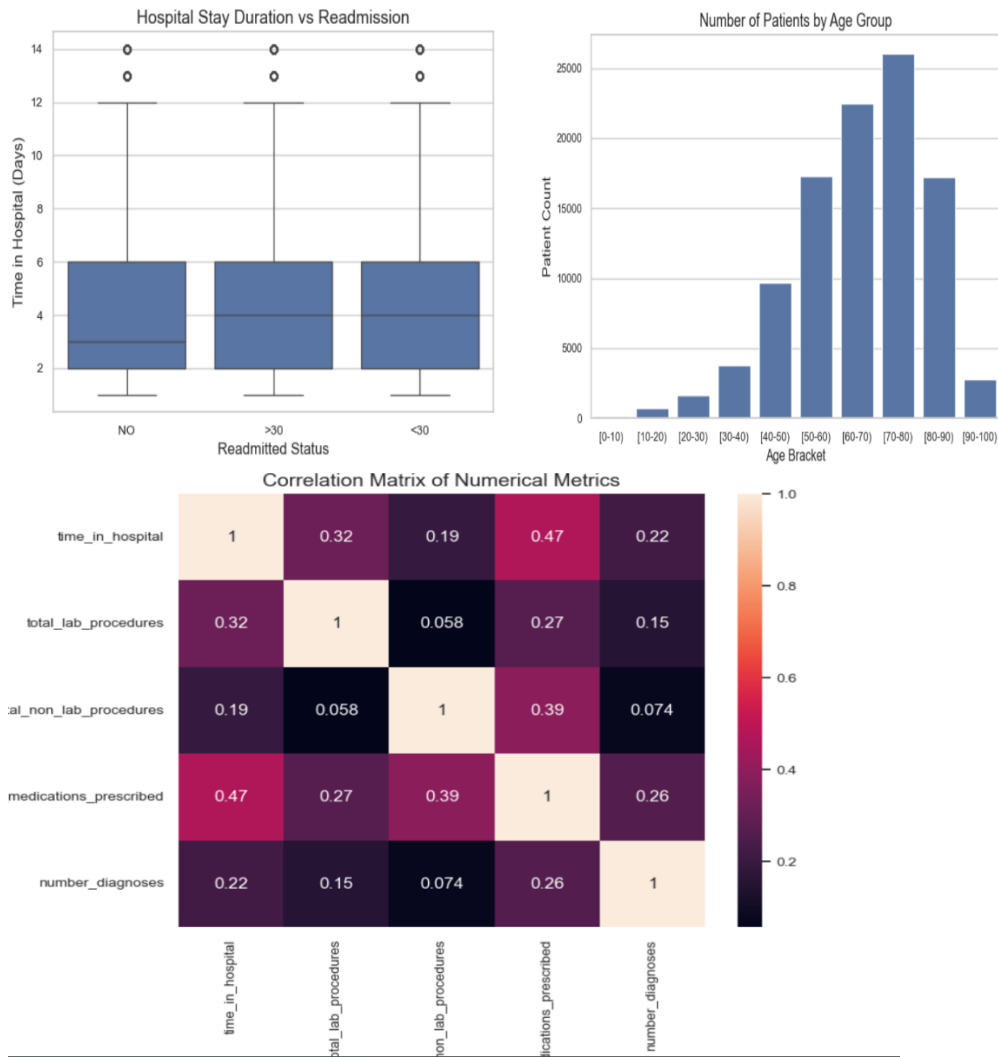
5. Key Insights

- **Demographic Risks:** Older age cohorts exhibited significantly longer hospital stay durations and a higher frequency of admissions compared to younger groups.
- **Clinical Utilization:** A strong positive correlation was observed between the number of lab procedures performed and the total number of medications prescribed during a patient's stay.
- **Readmission Patterns:** Patients with a higher count of pre-existing comorbidities and longer initial hospital stays showed a distinct increase in hospital readmission rates.
- **Medication Impact:** Specific shifts or frequent changes in diabetes medication regimes closely correlated with fluctuations in the time patients spent in the medical facility.

6. Visualizations

- **Box Plot:** Showing the distribution of "Time in Hospital" across different age groups to isolate stay durations and outliers.
- **Count Plot:** Displaying readmission counts segmented by gender and age cohort.

- **Heatmap:** Displaying the correlation matrix between numerical features like time in hospital, number of medications, and number of lab procedures.



7. Files Included

- diabetic_data.csv – Raw clinical dataset
- diabetic_data_analysis.ipynb – Jupyter Notebook containing the Python code for data cleaning, EDA, and visualizations
- README.md – Project description, objective summaries, and execution instructions

8. How to Use

1. Open `diabetic_data_analysis.ipynb` using Jupyter Notebook, JupyterLab, or VS Code.
2. Ensure the required Python libraries (pandas, numpy, matplotlib, seaborn) are installed in your environment.
3. Run the notebook cells step by step to observe the data cleaning steps, imputation methods, statistical summaries, and visual plots.
4. Modify the filtering parameters or chart aesthetics to explore additional clinical relationships if desired.

9. Conclusion

This project demonstrates how Python libraries such as Pandas, Matplotlib, and Seaborn can be effectively utilized to clean, process, and analyze complex healthcare data. The insights extracted from this analysis provide a clearer understanding of diabetic patient trends, which can assist medical institutions in optimizing patient monitoring, managing resources efficiently, and improving care strategies to mitigate readmission risks.